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Aircraft landing problems with aircraft classes

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Abstract This article focuses on the aircraft landing problem that is to assign landing times to aircraft approaching the airport under consideration. Each aircraft's landing time must be in a time interval encompassing a target landing time. If the actual landing time deviates from the target landing time additional costs occur which depend on the amount of earliness and lateness, respectively. The objective is to minimize overall cost. We consider the set of aircraft being partitioned into aircraft classes such that two aircraft of the same class are equal with respect to wake turbulence. We develop algorithms to solve the corresponding problem. Analyzing the worst case run-time behavior, we show that our algorithms run in polynomial time for fairly general cases of the problem. Moreover, we present integer programming models. We show by means of a computational study how optimality properties can be used to increase efficiency of standard solvers.

Keywords Aircraft scheduling · Landing times · Aircraft classes · Complexity · MIP models

1 Introduction

Runway complexes of major airports are very scarce and expensive resources and their capacity is gravely limited by separation requirements between aircraft, see Barnhart et al. (2003). If an arriving aircraft enters the radar range of an airport, then air traffic control must assign a landing time within a specific time window to this aircraft. In addition, a runway has to be assigned in case of multi-runway systems. Each arriving aircraft a has a preferred landing time or target landing time T_a, which results from the time of entry into the radar range and the preferred speed. The earliest landing time E_a can be reached with maximum speed. The latest landing time L_a is the latest time where the aircraft can land if it flies most fuel-efficient and is circling for the maximum allowable time.

When determining the landing times for succeeding aircraft, separation requirements have to be considered. A landing aircraft causes wake turbulence and, therefore, the following aircraft must have a safety distance. Heavy aircraft (wide-body jets) may generate more severe wake turbulence than a small aircraft. Therefore, landing a heavy aircraft, for example a Boeing 747, necessitates a large-time delay before another aircraft can land. The smaller the following aircraft is the more hazardous the air turbulence is. Hence, separation requirements depend on the weight class of succeeding aircraft, i.e., they are sequence-dependent. Usually, a limited number of aircraft classes is distinguished based on the maximum takeoff weight, see for example the separation standards of the International Civil Aviation Organisation (ICAO). Two kinds of separation requirements can be modeled: complete and successive separation. In the latter case, separation requirements only apply to aircraft landing consecutively on the same runway. When considering complete separation, separation requirements must be fulfilled for each pair of aircraft landing on the same runway.

The traditional sequencing policy at most airports is the first-come-first-served (FCFS) discipline. The aircraft landing problem (ALP) is to assign a landing time and a runway to

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each aircraft, and, hence, to construct a sequence of landing aircraft for each runway that minimizes the overall penalty for actual landing times deviating from target landing times. Each assigned landing time has to be in the corresponding time window and the separation requirements have to be fulfilled. Usually, approaches to solve the ALP can be adapted to incorporate takeoff operations.

The static version of the ALP assumes that the time window and the preferred landing times are known in advance for the whole planning horizon, see the reviews in Beasley et al. (2000) and Pinol and Beasley (2006). For an overview of approaches for the dynamic version we refer to Beasley et al. (2004) and Dear and Sherif (1991). Psaraftis (1978, 1980) developed a dynamic programming (DP) formulation of the ALP with successive separation for single runways with $E_a = 0$, i.e., all aircraft are available to land at time 0. Psaraftis (1980) considers classes of aircraft and assumes first, the separation requirements to be class-dependent and second, penalties to be non-decreasing in landing time. Under these assumptions, it is shown that the problem can be solved in time being polynomial in the number of aircraft but exponential in the number of classes. Bianco et al. (1999) extend the DP formulation to release dates $E_a = T_a \geq 0$ but still assume a latest landing time of $L_a = \infty$. Venkatakrishnan et al. (1993) extend the DP to the case of finite time windows $[E_a = T_a, L_a]$ and consider the case of multiple runways.

The most general mixed integer programming (MIP) formulations are presented in Beasley et al. (2000) and Pinol and Beasley (2006). They assume multiple runways, a finite time window $[E_a, L_a]$ with a target landing time $E_a \leq T_a \leq L_a$. The cost for landing aircraft a is linearly related to the deviation from T_a with possibly different penalties for earlier and later landings. Ernst et al. (1999) discuss the possibility to fix the order of two aircraft in a preprocessing step for the single runway case to speed up a Branch and Bound procedure. Pinol and Beasley (2006) also analyze nonlinear cost functions. Beasley et al. (2000) discuss a piecewise linear cost function with a single breakpoint. Soomer and Franx (2008) analyze airline-specific convex piecewise linear cost functions with multiple breakpoints. An extension to fairness aspects is considered in Soomer and Koole (2008). Balakrishnan and Chandran (2010) limit the number of overtaking manoeuvers to ensure some degree of fairness. They develop a DP approach for a single runway under this constrained position shifting constraint. Fahle et al. (2003) propose one MIP model with a continuous time axis and one MIP model with a discrete time axis. They compare both MIP models for single runway instances. A heuristic solution for a more detailed model of the terminal area is presented in Bianco et al. (2006). Even though most articles dealing with MIP formulations of the ALP are referring to Psaraftis (1980), they assume aircraft-dependent separation requirements instead of class-dependent separation requirements.

The article at hand contributes to this area in various ways. First, we determine the computational complexity of a reasonably general problem considering aircraft classes. We provide polynomial time algorithms for a fixed number of classes and, doing so, we considerably generalize results in Bianco et al. (1999), Psaraftis (1978, 1980), and Venkatakrishnan et al. (1993). In the most general setting we consider multiple runways, finite-time window $[E_a, L_a]$ with a target landing time $E_a \leq T_a \leq L_a$, and piecewise linear cost functions with multiple breakpoints. Second, we propose integer programming (IP) models to employ standard solvers. Third, we show that optimality properties developed for the problems at hand can be employed to design IP models such that efficiency of standard solvers can be increased.

The article is organized as follows. In Sect. 2, the problem at hand is formally defined. Furthermore, relations to other problems are outlined. In Sect. 3, we present polynomial time algorithms for the general problem with a fixed number of classes and special cases, respectively. Section 4 focuses on IP models and the use of standard solvers. Finally, in Sect. 5, conclusions and an outlook to further research are given.

2 Problem definition and relations to other problems

2.1 The aircraft landing problem with aircraft classes

We consider a set A of aircraft partitioned into the set $\mathcal{A} = \{A_1, \dots, A_W\}$ of W classes. Each class A_w, $w = 1, \dots, W$ contains n_w aircraft. Class $w(a)$ is the class aircraft a is contained in. Regarding real world applications, a reasonable partition can be found considering the sizes of aircraft, see for example the classification of ICAO.

Landing an aircraft occupies the corresponding runway for a specific time period. We refer to the end of this period as the landing time throughout this article. Accordingly, everything related to the position of a landing in the time horizon is defined with respect to the landing time. Note that, since we assume the length of the time period a landing occupies a runway to be known in advance, we easily could use the beginning of landing as a reference point as well.

Aircraft $a \in A$ has a target time T_a which reflects the ideal landing time for a. By slowing the aircraft down or speeding it up, the actual landing time may deviate from the target time. However, an earliest possible landing time E_a, $E_a \leq T_a$, and a latest possible landing time L_a, $L_a \geq T_a$ restrict the landing time to the interval $[E_a, L_a]$. Now, let $L_w^{\max}$ and $E_w^{\max}$ be a class-dependent upper bound for lateness $L_a - T_a$ and earliness $T_a - E_a$, respectively, with $w(a) = w$, that is

$$T_a - E_{w(a)}^{\max} \leq E_a \leq T_a \leq L_a \leq T_a + L_{w(a)}^{\max}$$

for each aircraft $a \in A$.

Consider two aircraft a and a' with $w(a) = w(a')$. We assume throughout the article that

- $E_a < E_{a'} \Longrightarrow T_a \leq T_{a'} \wedge L_a \leq L_{a'}$,
- $T_a < T_{a'} \Longrightarrow E_a \leq E_{a'} \wedge L_a \leq L_{a'}$, and
- $L_a < L_{a'} \Longrightarrow E_a \leq E_{a'} \wedge T_a \leq T_{a'}$.

Intuitively speaking, if there is an order of a and a' regarding one of earliest landing time, target time, and latest landing time, then the other ones must be ordered accordingly. We refer to this assumption as landing time window order. From the practical point of view, there are similar or identical aircraft within one class, so that there are similar speed and safety fuel regulations. Here, we may assume that an aircraft having earlier target time is closer to the airport. So, it most probably can land earlier, i.e., its earliest landing time is earlier. On the other hand, it has lower fuel level probably and, therefore it can keep flying shorter, i.e., its latest landing time is earlier. There are several cases in literature where special cases of the landing window order are assumed. In Psaraftis (1978, 1980), we have $E_i = T_i = 0$ and $L_i = \infty$. Harikiopoulo and Neogi (2011) assume $E_i = T_i$ and $L_i = \infty$. Although the model in Balakrishnan and Chandran (2010) is more general, the data the authors use in the proof of concept are according to our assumption since $E_i = T_i - e$ and $L_i = T_i + l$ hold where $l, e \geq 0$ are aircraft independent and T_i may be considered the estimated time of arrival.

For each class w, a piece-wise linear convex cost function $c_w(t) : [-E_w^{\max}, L_w^{\max}] \to \mathbb{R}$ reflects the additional cost depending on the deviation t of the actual landing time from the target time of an aircraft of class w. These additional costs may occur due to longer flight duration or speeding up the aircraft. We assume that $c_w(t) \geq 0$ for each $-E_w^{\max} \leq t \leq L_w^{\max}$, $c_w(0) = 0$, each cost function has a (dummy) break point at $t = 0$, and no cost function has more than K break points. W. l. o. g. we then can assume that c_w has exactly K break points. We address the kth, $1 \leq k \leq K$, class-specific break point by b_k^w and assume break points to be ordered such that $b_k^w \leq b_{k+1}^w$ for each w and $1 \leq k < K$. Finally, let k_w^* be the index of the break point at $t = 0$ for class w.

Furthermore, for each pair (w, w') of classes a minimum separation time $s_{w,w'}$ is given. We consider two types of requirements imposed by separation times. According to the first type, the minimum separation time $s_{w,w'}$ gives the minimum distance of the landing times of aircraft a and a' with $w(a) = w$ and $w(a') = w'$ which are scheduled to land consecutively on the same runway. We refer to this type of requirements as successive separation in the following. According to the second type, the minimum separation time $s_{w,w'}$ gives the minimum distance of the landing times of each pair of aircraft scheduled to land on the same runway. We refer to this type of requirements as complete separation in the following. For both types, separation times cover the period of time the succeeding aircraft occupies the runway and the minimum timespan between both landing operations. Thus, separation time $s_{w(a),w(a')}$ between aircraft a and a' covers minimum timespan between landing operations of types $w(a)$ and $w(a')$ and duration of landing operation of a'.

Finally, $\mathcal{R} = \{1, \ldots, R\}$ is a set of runways which are assumed to be identical and independent from each other. Throughout this article, we assume w. l. o. g. that all parameter values are integers and known in advance.

Definition 1 A schedule is a set $S \subset A \times \mathcal{R} \times \mathbb{N}$ such that there is exactly one tuple $(a, r, C_a) \in S$ for each $a \in A$.

The third value in a triple $(a, r, C_a) \in S$ is the landing time of the aircraft represented by the first value according to S. The second value represents the runway the aircraft is scheduled to land on according to S.

Definition 2 A schedule S is called feasible if

1. $E_a \leq C_a \leq L_a$ for each $(a, r, C_a) \in S$ and

2i. for each $(a, r, C_a) \in S$ and $(a', r, C_{a'}) \in S$ with $C_{a'} > C_a$ there is a $(a'', r, C_{a''}) \in S$ with $C_a < C_{a''} < C_{a'}$ or $C_{a'} \geq C_a + s_{w(a),w(a')}$ or

2ii. for each $(a, r, C_a) \in S$ there is no $(a', r, C_{a'}) \in S$ such that $a \neq a'$ and $C_a \leq C_{a'} < C_a + s_{w(a),w(a')}$.

Condition 1 ensures that a's landing time lies in the required time window. Alternative conditions 2i and 2ii represent successive separation and complete separation, respectively. Condition 2i assures that landing operations of aircraft a and a' must be separated by at least $s_{w(a),w(a')}$ if a is the immediate predecessor of a'. Condition 2ii states that landing operations of aircraft a and a' must be separated by at least $s_{w(a),w(a')}$ if a' is scheduled after a on r.

The focus is on successive separation throughout the article. In real world environments the interference of aircraft which are not scheduled consecutively is a rare event. Formally, both models are equivalent as long as the triangle inequality

$$s_{w,w'} + s_{w',w''} \geq s_{w,w''} \tag{1}$$

holds for separation times. Thus, we restrict the development of algorithms to successive separation. In Sect. 3.4, we sketch how to adapt the most general approach to complete separation.

Definition 3 Let $c(a, C_a) = c_{w(a)}(C_a - T_a)$ be the cost of landing aircraft a at time C_a and let $c(S) = \sum_{(a,r,C_a)\in S} c(a, C_a)$ be the cost of schedule S.

Definition 4 The aircraft landing problem with aircraft classes (ALP) is to find the minimum-cost feasible schedule.

Table 1 Notation

$a \in A$	Aircraft
$w \in W$	Aircraft classes
$\mathcal{A} = \{A_1, \ldots, A_W\}$	Partition of the set of aircraft into classes
n_w	Number of aircraft in class w
$w(a)$	Class aircraft a
E_a	Earliest landing time of a
T_a	Target landing time of a
L_a	Latest landing time of a
$L_w^{\max}$	Class dependent upper bound for lateness
$E_w^{\max}$	Class dependent upper bound for earliness
$c_w(t)$	Cost function
$1 \leq k \leq K$	Break points of the cost function
b_k^w	Class-specific break point
$s_{w,w'}$	Class-dependent minimum separation time
$\mathcal{R} = \{1, \ldots, R\}$	Set of runways
C_a	Assigned landing time
S	Schedule
$c(a, C_a)$	Cost of landing a at C_a

While many variants of aircraft landing problems have been considered in literature, only a few assume the set of aircraft under consideration to be separated into classes. In contrast to the article at hand, in Psaraftis (1978, 1980) all aircraft are assumed to be available to schedule at the beginning of the planning horizon and could be delayed infinitely long, i.e., $E_i = T_i = 0$ and $L_i = \infty$ is assumed. In Harikiopoulo and Neogi (2011), a special case of ALP has been considered where $E_i = T_i$ and $L_i = \infty$ are assumed. We considerably generalize both problem settings. Note also that in both problem settings the assumptions regarding the time windows and target landing times are more restrictive. For the aircraft landing problem without aircraft classes even minimizing the latest landing time is NP-hard on a single runway, see Psaraftis (1980). Note that the objective function of ALP is much more general. Many well-studied objective functions are special cases of our's, e. g., minimization of total tardiness ($\sum_a \max\{0, C_a - T_a\}$), which in turn generalizes minimization of maximum lateness ($\max_a \{C_a - T_a\}$), minimization of total completion time ($\sum_a C_a$), and minimization of the last landing time ($\max_a C_a$), see Pinedo (2012). In addition, our problem generalizes the basic setting which is NP-hard for individual aircraft, e. g., by incorporating multiple runways and landing time windows.

The used notation is summarized in Table 1.

2.2 Relation to other problems

Two fields in combinatorial optimization provide problem frameworks that are highly related to aircraft landing problems. In the following, we specify the relation and outline how results in the article at hand can be projected to the related problems.

Clearly, we can represent ALP in terms of machine-scheduling models as follows. Representing runways by machines and aircraft by jobs we obtain the following characteristics of the machine-scheduling model. We have R identical machines. Furthermore, we have families of jobs representing classes of aircraft. Jobs in the same family have equal processing times. Furthermore, family dependent and sequence dependent setup times are considered. Note that if job a' follows job a on the same machine then processing time of a' plus setup time from a's family to a''s family corresponds to separation time $s_{w(a),w(a')}$. Note also that usually, when family-dependent setup times are considered no setup occurs between consecutive jobs of the same family and often the setup time depends only on the family of the following job, see Potts and Kovalyov (2000) for a survey. In our problem, the setting is more general since setup times between jobs of the same family may be larger than zero and setup times depend on both jobs' families. Jobs have release dates, due dates, and target times which are independent except for the relation imposed by landing time window order. The objective is to minimize total cost where cost of scheduling a job is given as a function of the deviation of assigned completion time from target time according to c_w. To the best of our knowledge, this machine scheduling problem has not been considered yet. The complexity status seems to be open if the number of families is fixed.

However, we do know that the problem turns NP-hard for several generalizations. If we consider the number of families not to be fixed, then the problem is a generalization of the scheduling of jobs with equal processing times and sequence-dependent setup times on a single machine to minimize the makespan. This problem is equivalent to the traveling salesman problem and, thus, strongly NP-hard, see Pinedo (2012). If we instead assume processing times to be arbitrary, then the problem is a generalization of the scheduling of jobs with arbitrary processing times and due dates on a single machine to minimize maximum lateness which is strongly NP-hard, see Pinedo (2012).

Another field of combinatorial optimization ALP is related to is vehicle routing. Here, the problem setting covered by ALP is as follows. We consider a set of customers partitioned into subsets corresponding to the set of aircraft partitioned into classes. We assume that distances between customers only depend on the subset the customers are contained in. For example, we can think of a setting where customers are located in different cities and distances between two customers exclusively depend on the cities the customers are located in. Then, the distance between two customers corresponds to the separation time between two aircraft. Each customer has a service interval in which he has to be served and a target time in this interval when he prefers to be served. Lower bound and upper bound of the interval and the target time correspond to earliest landing time, latest landing time, and target time, respectively. Accordingly, we assume that a relation between these times according to landing time window order holds for each pair of customers located in the same city. Finally, runways correspond to vehicles we can dispatch. Now the objective is to serve each customer in his service interval such that total penalty for deviation of service time from target time is minimized. Again, this generalizes several well-studied objectives such as minimization of total travel distance and minimization of total deviation from service intervals. While the complexity for the problem with a fixed number of subsets seems to be open, the more general setting with an unbounded number of subsets is NP-hard as mentioned before.

3 Polynomial time algorithms

In the section at hand, we develop algorithms for ALP. These algorithms run in polynomial time for the special cases of ALP where the number of runways, as well as the number of aircraft classes are fixed. The most general variant of ALP we consider is the ALP with multiple runways and multiple aircraft classes. Special cases are derived by assuming either the number of runways or the number of classes or both to equal one. Clearly, as far as computational complexity is concerned only the polynomial algorithm for the most general case is needed. However, the algorithm runs in polynomial time, its actual run time may be very large when applied to real-world settings and this is why we provide more efficient algorithms for special cases. We develop two basic ideas for a single runway and multiple runways. These are applied to both, the problem with a single aircraft class and the problem with multiple aircraft classes.

First, we provide two basic optimality properties in Lemmas 1 and 2. These are employed when developing the algorithms in Sections 3.1–3.4.

Lemma 1 *Aircraft of each class w can be scheduled in non-decreasing lexicographic order of (E_a, T_a, L_a).*

Proof Consider a schedule S where $C_a > C_{a'}$, $w_a = w_{a'}$, and (E_a, T_a, L_a) is lexicographic smaller than $(E_{a'}, T_{a'}, L_{a'})$. Note that we must have $E_{a'} \le C_{a'} < C_a \le L_a$, then. We construct a schedule $\bar{S}$ by swapping a and a', that is, we set $\bar{C}_a = C_{a'}$ and $\bar{C}_{a'} = C_a$ where $\bar{C}_{a''}$ denotes the landing time of aircraft $a'' \in A$ according to $\bar{S}$. Clearly, $\bar{S}$ is feasible since

- separation times depend only on classes of consecutive aircraft and
- a and a' is landed in $[E_a, L_a]$ and $[E_{a'}, L_{a'}]$, respectively, since $\bar{C}_a = C_{a'} \ge E_{a'} \ge E_a$ and $\bar{C}_{a'} = C_a \le L_a \le L_{a'}$ hold due to landing time window order.

Since $c_{w(a)}$ is convex, both, $c(a, C_a)$ and $c(a', C_{a'})$ are convex as well. Since $T_a \le T_{a'}$, we have $c'(a, t) \ge c'(a', t)$ for each $E_{a'} \le t \le L_a$ where c' is the first derivative of c. Since $E_{a'} \le C_{a'} < C_a \le L_a$, swapping a and a' cannot increase the objective value. □

Note that Lemma 1 is not restricted to aircraft landing on the same runway. We refer to the order of aircraft according to lexicographic non-decreasing values of (E_a, T_a, L_a) as earliest landing time windows (ELW) rule. Accordingly, we assume that $A = \{1, \ldots, n\}$ such that each class is ordered in ELW with ties arbitrarily broken.

Definition 5 Given a schedule S a block is a set of aircraft B, $|B| = n'$, and a corresponding sequence σ such that

- all aircraft in B are scheduled on the same runway r,
- $C_{\sigma(k)} = C_{\sigma(k-1)} + s_{w(\sigma(k-1)),w(\sigma(k))}$ for each $2 \le k \le n'$,
- $\sigma(1)$ is the first aircraft on r or $C_{\sigma(1)} > C_j + s_{w(j),w(\sigma(1))}$ where j is the immediate predecessor of $\sigma(1)$ on r, and
- $\sigma(n')$ is the last aircraft on r or $C_j > C_{\sigma(n')} + s_{\sigma(n'),w(j)}$ where j is the immediate successor of $\sigma(n')$ on r.

Definition 6 An aircraft a is a fixed aircraft of type $\alpha \in \{0, 1, \ldots, K, K+1\}$ if

- $\alpha = 0$ and $C_a = E_a$,
- $\alpha \in \{1, \ldots, K\}$ and $C_a - T_a$ equals the αth break point of $c_{w(a)}$, or
- $\alpha = K + 1$ and $C_a = L_a$.

Lemma 2 *There is an optimal schedule such that for each block B there is an aircraft a, $a \in B$, such that a is a fixed aircraft of type $\alpha \in \{0, 1, \ldots, K, K+1\}$.*

Proof Let S, B, and C_a be an optimal schedule, a block according to S, and the landing time of aircraft a according to S, respectively. Suppose there is no fixed aircraft in B. Let $\bar{c}_a$ be the marginal cost of aircraft a in C_a for each $a \in B$. Note that $\bar{c}_a$ is unique for each $a \in B$ since each $a \in B$ is not fixed. Let

$$c_B^d = \sum_{a \in B} \bar{c}_a$$

be the marginal cost of delaying B, that is simultaneously delaying all aircraft in B. Since no aircraft in B is fixed, there is a sufficiently small ϵ, $\epsilon > 0$, by which we can delay and accelerate, respectively, B without

- any aircraft a's landing time C_a reaching E_a or L_a or
- $C_a - T_a$ reaching any break point of $c_{w(a)}$.

Hence, if $c_B^d < 0$ or $c_B^d > 0$, then we delay or accelerate, respectively, B by ϵ and obtain a better schedule which is a contradiction to the assumption. Thus, $c_B^d = 0$. Then, we can delay or accelerate, respectively, B such that

- there is an aircraft $a \in B$ such that a is a fixed aircraft of type $\alpha \in \{0, 1, \ldots, K, K+1\}$ or
- B merges with another block B' and we obtain a new block $B'' = B \cup B'$.

Note that blocks can be merged no more than $n-1$ times before we end up with a single block. Delaying or accelerating the single remaining block will lead to a fixed aircraft a of type $\alpha \in \{0, 1, \ldots, K, K+1\}$ with the reasoning above. □

3.1 Single class on a single runway

In the section at hand, we design an algorithm to solve ALP if $W = R = 1$. Due to Lemma 1 we can model this special case as a linear programming (LP) problem since we have a unique order of all aircraft. Using continuous variables for each aircraft representing the landing times we can easily ensure that no separation constraint is violated. Additional variables are needed to accurately represent the cost function. It seems that we need at least $O(Kn)$ variables. Then, the run time complexity of our algorithm is smaller than the complexity for solving the LP problem.

The basic idea is to have a graph where edges represent the schedule of aircraft between two fixed aircraft. We show how to design the structure of the graph in the proof of Theorem 1. Lemma 3 serves as a building block stating that we can compute lengths of arcs in the graph mentioned above in polynomial time. The corresponding method is provided in the proof of Lemma 3. In this section, for the sake of shortness we let $s = s_{1,1}$.

Lemma 3 *If $R = W = 1$, aircraft a and $a', a < a'$, are fixed aircraft of type α and α′, respectively, and each aircraft $a'', a < a'' < a'$, is in the same block as a or is in the same block as a', then we can find the optimal schedule and corresponding cost for the subset $\{a+1, \ldots, a'\}$ of aircraft in $O(n)$.*

Proof Clearly, the landing time of aircraft a and a' can be considered fixed since they are assumed to be fixed aircraft of type α and α', respectively. Considering Lemma 1, we can assume that aircraft is scheduled in ELW. Let B and B' be the block a and a', respectively, is contained in. Note that, possibly, $B = B'$. Since there is no further block consisting of a subset of aircraft $a+1, \ldots, a'-1$ we have a single idle time interval of length

$$t = C_{a'} - C_a - (a' - a)s$$

if $t > 0$. We then set

$$C_{a''} = C_{a''-1} + s \text{ for each } a'' \in \left\{a+1, \ldots, a'-1\right\}, \quad (2)$$

that is the single idle time period occurs immediately before aircraft a'. We can check feasibility and evaluate the corresponding schedule in $O(n)$. Now, we modify the schedule by increasing the landing time of the last aircraft a'' in B by t and, therefore, setting $B = B \setminus \{a''\}$ and $B' = B' \cup \{a''\}$. Now, we can check the feasibility and evaluate the modified schedule in $O(1)$ since the landing time of exactly one aircraft has been modified. After applying this modifying procedure $O(n)$ times, we derive all schedules with a single idle time period of length t. Minimum total cost is then found in $O(n)$ time among feasible schedules. If there is no feasible schedule, then we set total cost to ∞.

If $t = 0$, there is no idle time period and $B = B'$. Then, completion times are set according to (2) and we can check feasibility and derive total cost in $O(n)$. If the schedule is not feasible, then we set total cost to ∞.

If $t < 0$, then no feasible schedule exists. Accordingly, we set total cost to ∞. □

Clearly, we can find an optimal schedule for a subset of aircraft in linear time as well in the following two cases.

- If aircraft a is a fixed aircraft of type α and each aircraft a', $a' < a$, is in the same block as a, then we can find the only schedule for aircraft $1, \ldots, a$ fulfilling these prerequisites by setting

  $$C_{a'} = C_{a'+1} - s \text{ for each } a' \in \{1, \ldots, a-1\} \quad (3)$$

  since C_a can be assumed to be fixed. Then, checking feasibility and evaluating is straightforward.
- If aircraft a is a fixed aircraft of type α and each aircraft $a', a' > a$, is in the same block as a, then we can find the only schedule for aircraft $a+1, \ldots, n$ fulfilling these prerequisites by setting

  $$C_{a'} = C_{a'-1} + s \text{ for each } a' \in \{a+1, \ldots, n\} \quad (4)$$

  since C_a can be assumed to be fixed. Checking feasibility and evaluating is straightforward.

Theorem 1 *ALP can be solved in $O(K^2 n^3)$ if $R = W = 1$.*

Proof We define a weighted directed acyclic graph (DAG) $G = (V, E, l)$ as follows. The set of nodes V encompasses a dummy source node (0), a dummy sink node $(n+1)$, and a node (a, α) for each $a \in A$ and $0 \leq \alpha \leq K+1$. Node (a, α) represents aircraft a being a fixed aircraft of type α.

There are arcs $((0), (a, \alpha))$ and $((a, \alpha), (n+1))$ for each $a \in A$ and $0 \leq \alpha \leq K+1$. Arc $((0), (a, \alpha))$ represents aircraft a being the first fixed aircraft in the schedule and a being a fixed aircraft of type α. Note that we can assume aircraft a', $a' < a$, to be in the same block as a due to Lemma 2. Arc $((a, \alpha), (n+1))$ represents aircraft a being the last fixed aircraft in the schedule and a being a fixed aircraft of type α. We assume aircraft a', $a' > a$, to be in the same block as a due to Lemma 2. Length of arc $((0), (a, \alpha))$ and arc $((a, \alpha), (n+1))$ is determined using (3) and (4), respectively, if the corresponding partial schedule is feasible and is set to ∞ otherwise.

Furthermore, there is an arc $((a, \alpha), (a', \alpha'))$ if $a < a'$. Arc $((a, \alpha), (a', \alpha'))$ represents a and a' being a fixed aircraft of type α and α', respectively, and each aircraft a'', $a < a'' < a'$, may not be a fixed aircraft. Then, we can assume aircraft a'', $a < a'' < a'$, to be in a's block or in a''s block (possibly both) due to Lemma 2. Length of arc $((a, \alpha), (a', \alpha'))$ is determined according to Lemma 3. Note that it can be seen easily that G is a DAG since no arcs lead to (0) or start in $(n+1)$ and for each arc $((a, \alpha), (a', \alpha'))$ we have $a < a'$.

Finding a shortest path from (0) to $(n+1)$ corresponds to finding a minimum cost schedule. Clearly, $|V| \in O(Kn)$ and $|E| \in O(K^2n^2)$. Since cost of each edge can be computed in $O(n)$ according to Lemma 3 and using (3) and (4) constructing the graph takes $O(K^2n^3)$ while the shortest path can be found in $O(K^2n^2)$. Hence the problem can be solved in $O(K^2n^3)$. □

Fig. 1 provides graph G for a problem instance with ten aircraft and $K = 1$. Hence, there are three nodes (a, α), $\alpha \in \{0, 1, 2\}$, corresponding to each aircraft $a \in A$. We depict one path by solid arcs. Some of the remaining arcs are drawn as dashed arcs. Note that the arcs shown in Fig. 1 are picked as examples. In fact, there exists an arc from (a, α) to (a', α') if $a < a'$. The path can be interpreted as follows. Aircraft 2 is a fixed aircraft of type 0, that is, $C_2 = E_2$, aircraft 6 is a fixed aircraft of type 1, that is, $C_6 - T_6$ equals the (unique) break point of c_1, and aircraft 9 is a fixed aircraft of type 2, that is, $C_9 = L_9$.

Fig. 2 provides a schedule corresponding to the path shown in Fig. 1. While we restrict ourselves to separation requirements between landing times in our analysis we break the separation time down into landing duration and idle time between landing operations in Fig. 2 to support an intuitive understanding. Consequently, white rectangles represent the landing operations of the corresponding aircraft. Grey rectangles represent the minimum idle time between landing operations of two consecutive aircraft in the same block. The landing times of aircraft 2, 6, and 9 are outlined. We have three blocks consisting of aircraft 1–3, 4–8, and 9–10, respectively.

Obviously, landing times of aircraft 2, 6, and 9 (b_1 represents the unique break point of c_1) can be directly obtained from the path depicted in Fig. 1. Note, however, that landing times of remaining aircraft cannot be identified from the path without further information. In fact, there may be more than one schedule having 2, 6, and 9 as fixed aircraft of type 0, 1, and 2, respectively. For example, aircraft 8 landing in $C_9 - s$ and keeping all other landing times would yield a schedule corresponding to the path, as well. Nevertheless, we assume that landing time of aircraft 8 as shown in Fig. 2 is obtained by applying Lemma 3 and, therefore, is optimal given that 2, 6, and 9 is a fixed aircraft of type 0, 1, and 2, respectively.

3.2 Multiple classes on a single runway

In the section at hand, we generalize the basic idea used in Section 3.1 to solve ALP with multiple aircraft classes and a single runway.

Lemma 4 *If*

- $R = 1$,
- *aircraft a and a' are fixed aircraft of type α and α', respectively,*
- *a is scheduled earlier than a', that is, $C_a < C_{a'}$,*
- *a (possibly empty) subset of aircraft A'_w for each class $w = 1, \ldots, W$ is scheduled between a and a', and*
- *each aircraft scheduled between a and a' is in a's block or in a''s block*

then we can find an optimal schedule for aircraft in $A' = \bigcup_{w=1}^{W} A'_w \cup \{a'\}$ and corresponding cost in $O\left(W^3 n^{2W^2}\right)$.

Proof Clearly, the landing time of aircraft a and a' can be considered fixed since they are assumed to be fixed aircraft of type α and α', respectively. Considering Lemma 1, we can assume that aircraft of class w, $1 \leq w \leq W$, are scheduled in ELW order.

We design a DP approach which step by step either

- schedules an aircraft at the end of those aircraft which have been scheduled after a without idle time but separation time or
- schedules an aircraft immediately before those aircraft which have been scheduled before a' without idle time but separation time.

Following this idea, two sequences of aircraft are constructed. The first one is constructed in forward direction such that the first separation time period starts at C_a (which is given since a is a fixed aircraft of type α). The second one is constructed in

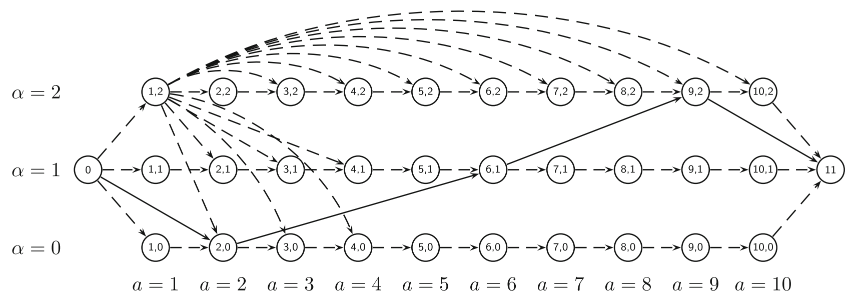

Fig. 1 Example for graph G with $n = 10$ and $K = 1$

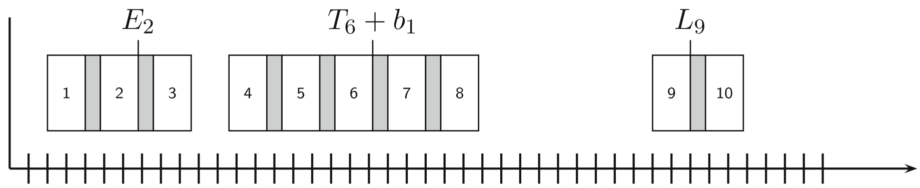

Fig. 2 Example for schedule

backward direction such that the last separation time period which is associated to aircraft a' ends at $C_{a'}$ (which is given since a' is a fixed aircraft of type α').

Now consider node (k^f, k^b, w^f, w^b) where $k^f, k^b \in \mathbb{N}^{W\times W}$ and $w^f, w^b \in \{1, \ldots, W\}$. Tuple k^f and k^b give the number $k^f_{w,w'}$ and $k^b_{w,w'}$ of separation time periods from type w to type w' that have been scheduled in the forward sequence and in the backward sequence, respectively. Type w^f and w^b give the type of the last aircraft and the first aircraft in the forward sequence and the backward sequence, respectively. We consider a node (k^f, k^b, w^f, w^b) for each $w^f, w^b \in \{1, \ldots, W\}$ and $k^f_{w,w'}, k^b_{w,w'} \in \left\{0, \ldots, \min\left\{|A'_w|, |A'_{w'}|\right\}\right\}$ such that $k^f_{w,w'} + k^b_{w,w'} \le \min\left\{|A'_w|, |A'_{w'}|\right\}$.

A source node and a dummy sink node are given by $(0, \ldots, 0, w(a), w(a'))$ and $(n, \ldots, n, W+1, W+1)$, respectively. Now, there is an arc from (k^f, k^b, w^f, w^b) to non-sink node $(\bar{k}^f, \bar{k}^b, \bar{w}^f, \bar{w}^b)$ if

- transition from (k^f, k^b, w^f, w^b) to $(\bar{k}^f, \bar{k}^b, \bar{w}^f, \bar{w}^b)$ represents scheduling an aircraft a'' of type $\bar{w}^f$ which is not scheduled yet at the end of the forward sequence, that is
  - $\sum_{w'=1}^{W} \left(k^f_{w',\bar{w}^f} + k^b_{\bar{w}^f,w'}\right) < |A'_{\bar{w}^f}|$,
  - $\bar{k}^f_{w,w'} = k^f_{w,w'}$ for $(w, w') \neq (w^f, \bar{w}^f)$
  - $\bar{k}^f_{w^f,\bar{w}^f} = k^f_{w^f,\bar{w}^f} + 1$
  - $k^b = \bar{k}^b$, and
  - $w^b = \bar{w}^b$,

  or
- transition from (k^f, k^b, w^f, w^b) to $(\bar{k}^f, \bar{k}^b, \bar{w}^f, \bar{w}^b)$ represents scheduling an aircraft a'' of type $\bar{w}^b$ which is not scheduled yet at the beginning of the backward sequence, that is
  - $\sum_{w'=1}^{W} \left(k^f_{w',\bar{w}^f} + k^b_{\bar{w}^f,w'}\right) < |A'_{\bar{w}^b}|$,
  - $k^f = \bar{k}^f$,
  - $w^f = \bar{w}^f$,
  - $\bar{k}^b_{w,w'} = k^b_{w,w'}$ for $(w, w') \neq (\bar{w}^b, w^b)$ and
  - $\bar{k}^b_{\bar{w}^b,w^b} = k^b_{\bar{w}^b,w^b} + 1$.

For both branches, the first condition ensures that not all aircraft of type $\bar{w}^f$ and $\bar{w}^b$, respectively, have been scheduled already. The remaining conditions consider the modification of counters and types. The length of this arc equals the cost of the represented scheduling of aircraft a'' if $E_{a''} \le C_{a''} \le L_{a''}$. This can easily be checked since C_a and $C_{a'}$ are fixed and, consequently,

$$C_{a''} = C_a + \sum_{w=1}^{W} \sum_{w'=1}^{W} k^f_{w,w'} s_{w,w'} \text{ and}$$

$$C_{a''} = C_{a'} - \sum_{w=1}^{W} \sum_{w'=1}^{W} k^b_{w,w'} s_{w,w'}$$

if a'' is scheduled in the forward sequence and in the backward sequence, respectively. Analogously, deriving scheduling cost for a'' can be done. If $E_{a''} > C_{a''}$ or $L_{a''} < C_{a''}$, then the length of the corresponding arc is ∞.

Finally, there is a transition from (k^f, k^b, w^f, w^b) to sink node $(n, \ldots, n, W+1, W+1)$ if

- $\sum_{w'=1}^{W} \left(k^f_{w',w} + k^b_{w,w'}\right) = |A'_w|$ for each $w \in \{1, \ldots, W\}$ and
- $\sum_{w=1}^{W} \sum_{w'=1}^{W} s_{w,w'} \left(k^f_{w,w'} + k^b_{w,w'}\right) + s_{w^f,w^b} \le C_{a'} - C_a$.

The first condition ensures that each aircraft in A' is scheduled. The second condition checks whether total separation time does not exceed the length of time interval $[C_a, C_{a'}]$.

We have $O\left(W^2 n^{2W^2}\right)$ nodes and, therefore, $O\left(W^3 n^{2W^2}\right)$ arcs and transitions, respectively. Note that the resulting graph is acyclic since no transition leads to the source node, no transition starts at the sink node and for each transition only involving other nodes, one of the counters in k^f or k^b is strictly increasing. Hence, we can find the optimal schedule for A' in $O\left(W^3 n^{2W^2}\right)$ time. If there is no path from $(0, \ldots, 0, w(a), w(a'))$ to $(n, \ldots, n, W+1, W+1)$, then there is no feasible schedule for A' and we set total cost to ∞. $\square$

Lemma 5 *If $R = 1$, aircraft a is a fixed aircraft of type α, the subset of aircraft $A'_w, w = 1, \ldots, W$, is scheduled before a, and all aircraft in $A'_w, w = 1, \ldots, W$, are in a's block, then we can find an optimal schedule for aircraft in $A' = \bigcup_{w=1}^{W} A'_w \cup \{a\}$ and corresponding cost in $O\left(W^2 n^{W^2}\right)$.*

Proof The idea is the same as in the proof for Lemma 4. However, only the backward sequence is considered here. Then, we have $O\left(W n^{W^2}\right)$ nodes and, therefore, $O\left(W^2 n^{W^2}\right)$ arcs. Hence, we can find the optimal schedule for A' in $O\left(W^2 n^{W^2}\right)$ time. $\square$

Lemma 6 *If $R = 1$, aircraft a is a fixed aircraft of type α, the subset of aircraft $A'_w, w = 1, \ldots, W$, is scheduled after a, and all aircraft in $A'_w, w = 1, \ldots, W$, are in a's block, then we can find an optimal schedule for aircraft in $A' = \bigcup_{w=1}^{W} A'_w$ and corresponding cost in $O\left(W^2 n^{W^2}\right)$.*

Proof The idea is the same as in the proof for Lemma 4. However, only the forward sequence is considered here. Then, we have $O\left(W n^{W^2}\right)$ nodes and, therefore, $O\left(W^2 n^{W^2}\right)$ arcs. Hence, we can find the optimal schedule for A' in $O\left(W^2 n^{W^2}\right)$ time. $\square$

Theorem 2 *ALP can be solved in $O(K^2 W^5 n^{2W^2+2W})$ if $R = 1$.*

Proof We define a weighted DAG $G = (V, E, c)$ as follows. The set of nodes V encompasses a dummy source node (0), a dummy sink node $(n+1)$, and a node $(w', \alpha, k_1, \ldots, k_W)$ for each $1 \leq w' \leq W$, $0 \leq \alpha \leq K+1$, and $0 \leq k_w \leq n_w$, $1 \leq w \leq W$. Let a be the $k_{w'}$th aircraft of class w' in ELW order. Node $(w', \alpha, k_1, \ldots, k_W)$ represents a being a fixed aircraft of type α and k_w aircraft of class w landing not later than C_a for each $1 \leq w \leq W$.

There are arcs $((0), (w', \alpha, k_1, \ldots, k_W))$ and $((w', \alpha, k_1, \ldots, k_W), (n+1))$ for each node $(w', \alpha, k_1, \ldots, k_W)$ being neither source node nor sink node. Arc $((0), (w', \alpha, k_1, \ldots, k_W))$ represents aircraft a being the first fixed aircraft in the schedule, a being a fixed aircraft of type α, and k_w aircraft of class w landing not later than C_a. Arc $((w', \alpha, k_1, \ldots, k_W), (n+1))$ represents aircraft a being the last fixed aircraft in the schedule, a being a fixed aircraft of type α, and $n_w - k_w$ aircraft of class w landing after C_a. Since aircraft of each class can be assumed to be in ELW order, we can derive the subsets A'_w of aircraft from k_w for both types of arcs. Length of arc $((0), (w', \alpha, k_1, \ldots, k_W))$ and $((w', \alpha, k_1, \ldots, k_W), (n+1))$ is derived according to Lemmas 5 and 6.

Now, we consider arcs where neither the source node nor the sink node is involved. There is an arc $((w', \alpha, k_1, \ldots, k_W), (\bar{w}', \bar{\alpha}, \bar{k}_1, \ldots, \bar{k}_W))$ if

- $C_a < C_{\bar{a}}$ holds for corresponding a and $\bar{a}$ where
- $\bar{k}_w \geq k_w$ for each $1 \leq w \leq W$, and
- $\bar{k}_{\bar{w}'} > k_{\bar{w}'}$

where a and $\bar{a}$ is the $k_{w'}$th aircraft of class w' and the $\bar{k}_{\bar{w}'}$th aircraft of class $\bar{w}'$, respectively, in ELW order.

Arc $((w', \alpha, k_1, \ldots, k_W), (\bar{w}', \bar{\alpha}, \bar{k}_1, \ldots, \bar{k}_W))$ represents a and $\bar{a}$ being a fixed aircraft of type α and $\bar{\alpha}$, respectively, $\bar{k}_w - k_w$ aircraft of class w landing after C_a and not after $C_{\bar{a}}$, and each aircraft a', $C_a < C_{a'} < C_{\bar{a}}$, may not be a fixed aircraft. Due to Lemma 1 we can assume that the corresponding aircraft of class w is aircraft $k_w + 1$ to $\bar{k}_w$ in ELW order for each $1 \leq w \leq W$. Length of each arc is determined according to Lemma 4. Again, the graph obtained is obviously a DAG since counters are non-decreasing and at least one of them is strictly increasing on each arc.

Finding a shortest path from (0) to $(n+1)$ corresponds to finding a minimum cost schedule. Clearly, $|V| \in O(K W n^W)$ and $|E| \in O(K^2 W^2 n^{2W})$. Since cost of each edge can be computed in $O\left(W^3 n^{2W^2}\right)$ due to Lemmas 4–6 constructing the graph takes $O(K^2 W^5 n^{2W^2+2W})$ while the shortest path can be found in $O(K^2 W^2 n^{2W})$. Hence the problem can be solved in $O(K^2 W^5 n^{2W^2+2W})$. $\square$

3.3 Single class on multiple runways

In this section, we let $s = s_{1,1}$. For technical reasons, we break down s into required runway idle time s' between two landing operations and landing duration as outlined in Section 2.1. However, s' is used for initialization issues only.

Due to Lemma 2 we assume that $C_a \in \mathcal{T}$ where

$$\mathcal{T} = \left\{T_a + b_k^a + l \cdot s \mid a \in A, 0 \leq k \leq K+1, -n \leq l \leq n\right\}$$

with b_k^a, $1 \leq k \leq K$, is the kth break point of $c_{w(a)}$, $b_0^a = E_a - T_a$, and $b_{K+1}^a = L_a - T_a$.

Next, we adapt resource profiles developed in Baptiste (2000) for our purposes.

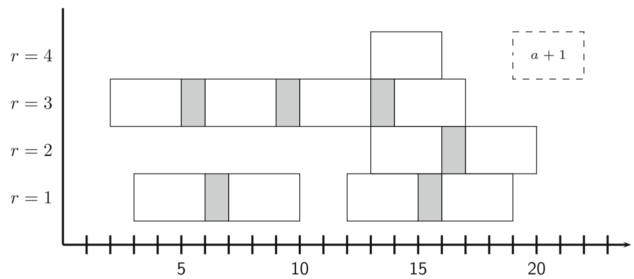

Fig. 3 Example for schedule on four runways

Definition 7 A runway occupation profile (ROP) is a vector $(O_1, \ldots, O_R)$, $O_R \geq \ldots \geq O_1 \geq -s'$, $O_r \in \mathcal{T}$ for each $1 \leq r \leq R$, $O_R - O_1 \leq s$.

A ROP $(O_1, \ldots, O_R)$ specifies that all runways are occupied by landing operations before O_1, exactly r, $1 \leq r \leq R-1$, runways are not occupied by landing operations in timespan $[O_r, O_{r+1}]$, and R runways are not occupied by landing operations starting at O_R. Note that we do not have to distinguish between runways since they are assumed to be identical. Therefore, it is sufficient to keep track of the number of available runways at a certain point of time. Note that, furthermore, given a ROP no further aircraft can land such that its landing is completed before $O_1 + s$.

Fig. 3 sketches a schedule for aircraft of a single type on four runways. Again, we break the separation time down into landing duration and idle time between landing operations. Each white rectangle represents the landing operation of an aircraft and, thus, for each aircraft the right bound of the corresponding rectangle marks its landing time. A gray block represents the minimum required period of time between two consecutive landing operations. Hence, in this example we have $s' = 1$ and $s = 4$. So, assuming that the aircraft depicted as solid lines in Fig. 3 are aircraft $1, \ldots, a$ the corresponding ROP are (16, 17, 19, 20). Thus, the earliest possible landing time for aircraft $a+1, \ldots, n$ is $16 + 4 = 20$.

Note that once we decided to schedule aircraft $1, \ldots, a$ such that a's landing is completed at C_a, then we can assume that no runway is available before $C_a - s$ for aircraft $a+1, \ldots, n$ due to Lemma 1.

Now, assume we schedule aircraft $a+1$ depicted as dashed lines in Fig. 3, that is, we set $C_{a+1} = 22$. Note that we can assume that we always schedule aircraft $a+1$ on the runway becoming available earliest. Scheduling $a+1$ modifies the ROP to (18, 19, 20, 22) for the following reasons. Clearly, runway 4 is occupied until 22 and no other runway is occupied for that long. Therefore, $O_4 = 22$. Also 1 and 2 are occupied until 19 and 20, respectively. Since runway 3 is available before 19 we have $O_2 = 19$ and $O_3 = 20$. Runway 3 becomes available at 17. However, due to Lemma 1 we can assume that no aircraft $a+2, \ldots, n$ completes its landing on runway 3 earlier than 22. This fact can be reflected by assuming that runway 3 does not become available before 18.

The number of ROP is bounded as follows. Of course, O_1 can take $|\mathcal{T}| \in O\left(Kn^2\right)$ values. Since $O_R \geq \ldots \geq O_1 \geq \max\{-s', O_R - s\}$ there are $O(Kn)$ possible values for each O_r, $2 \leq r \leq R$. Thus, there are $O\left(K^R n^{R+1}\right)$ ROPs.

Theorem 3 *ALP can be solved in* $O(RK^{R+1}n^{R+4})$ *if* $W = 1$.

Proof We define a weighted DAG $G = (V, E, c)$ as follows. The set of nodes V encompasses a source node $(0, (-s', \ldots, -s'))$, a dummy sink node $(n+1)$, and a node (a, rop) for each $a \in A$ and rop being a ROP. Node (a, rop) represents aircraft $1, \ldots, a$ being scheduled and runways being occupied as specified by rop.

Next, consider the transformation of ROP rop into ROP rop' by scheduling aircraft $a+1$. The basic idea of the algorithm is to schedule aircraft $1, \ldots, n$ according to ELW in increasing order. Therefore, we assume that aircraft $1, \ldots, a$ have been scheduled already and $C_{a'} \geq C_{a'-1}$, $2 \leq a' \leq a$. We say that runway profile $(O_1, \ldots, O_R)$ can be transformed into runway profile $(O'_1, \ldots, O'_R)$ with respect to aircraft $a+1$ if the following conditions hold.

- For O'_R we must have $L_{a+1} \geq O'_R \geq \max\{O_1 + s, E_{a+1}\}$. Clearly, the point of time the last runway becomes available is the landing time of $a+1$, that is, $O'_R = C_{a+1}$. Note that each aircraft a', $a' \leq a$ must have $C_{a'} \leq C_{a+1}$ due to Lemma 1. Since landing of aircraft $a+1$ cannot be finished before $\max\{O_1 + s, E_{a+1}\}$ or after L_{a+1} the condition must be fulfilled.
- If $O_r \leq O'_R - s < O_{r+1}$, $1 \leq r \leq R-1$, then $O'_{r'} = O'_R - s$ for each $r' \leq r-1$ and $O'_{r'} = O_{r'+1}$ for each $r \leq r' \leq R-1$. If $O_R \leq O'_R - s$, then $O'_{r'} = O'_R - s$ for each $r' \leq R-1$. Each runway is assumed to be occupied before $C_{a+1} - s = O'_R - s$ since no aircraft a', $a' > a+1$ lands earlier than $a+1$ due to Lemma 1. In addition, for each point of time t, $C_{a+1} - s < t < O'_R$, the number of available runways is reduced by one.

We denote $rop \overset{a}{\Rightarrow} rop'$ if ROP rop can be transformed into ROP rop' with respect to aircraft a.

We have two types of arcs. First, for two nodes $(a-1, rop)$ and (a, rop') there exists an arc $((a-1, rop), (a, rop'))$ if $a \in A$ and $rop \overset{a}{\Rightarrow} rop'$. Second, there is a node $((n, rop), (n+1))$ for each rop. It remains to show how to set weights. The second type of arcs has weight zero. The weights for the first type of arc reflect the cost implied by landing aircraft a at O'_R according to rop'. Clearly, a path from $(0, (-s', \ldots, -s'))$ to $(n+1)$ represents a feasible schedule and the shortest path outlines the minimum cost schedule. The constructed graph is acyclic due to the definition of transformability of one ROP into an other.

We have $O(K^R n^{R+2})$ nodes. Each node is connected to no more than $O(Kn^2)$ nodes due to the second compatibility condition which states that given a ROP rop choosing the landing time of aircraft a fully specifies the resulting ROP rop'. Thus, the number of arcs is in $O(K^{R+1}n^{R+4})$. Note that constructing each arc takes $O(R)$ time. Then, we can solve the problem in $O(RK^{R+1}n^{R+4})$ time. □

3.4 Multiple classes on multiple runways

Due to Lemma 2 we assume that $C_a \in \mathcal{T}'$ where

$$\mathcal{T}' = \left\{ T_a + b_k^a + \sum_{w=1}^{W} \sum_{w'=1}^{W} l_{w,w'} \cdot s_{w,w'} \mid a \in A, 0 \leq k \leq K+1, -n \leq l_{w,w'} \leq n, 1 \leq w, w' \leq W \right\}$$

with $b_k^a, 1 \leq k \leq K$, is the kth break point of $c_{w(a)}$, $b_0^a = E_a - T_a$, and $b_{K+1}^a = L_a - T_a$. We generalize ROPs introduced in Sect. 3.3 for our purposes.

Definition 8 A general ROP is a vector $((O_1, w_1), \ldots, (O_R, w_R))$ where

- $1 \leq w_r \leq W$ and $O_r \in \mathcal{T}'$ or
- $w_r = -1$ and $O_r = -1$

for each $1 \leq r \leq R$.

Here, if $1 \leq w_r \leq W$ and $O_r \in \mathcal{T}'$, $1 \leq r \leq R$, then O_r is the landing time of aircraft a scheduled last on runway r so far. Note that $O_r \in \mathcal{T}'$ can be assumed due to Lemma 2. Furthermore, $w_r = w(a)$. If $w_r = O_r = -1$, $1 \leq r \leq R$, then no aircraft has been scheduled on runway r so far. Hence, we have no more than

$$O\left(W^R \left(|\mathcal{T}'|\right)^R\right) = O\left(W^R \left(Knn^{W^2}\right)^R\right) = O\left(W^R K^R n^{R(W^2+1)}\right)$$

general ROPs. The basic idea of our approach is to schedule aircraft on each runway in a forward direction. Hence, once we decided to schedule aircraft a such that a's landing is completed at C_a on runway r, then we can assume that r is not available before C_a for any aircraft not scheduled yet. Furthermore, we make use of ELW order for each class.

Theorem 4 *ALP can be solved in* $O\big(RW^{R+1}K^{R+1} \cdot n^{(R+1)W^2+W+R+1}\big)$.

Proof We define a weighted DAG $G = (V, E, c)$ as follows. The set of nodes V encompasses a source node $(0, \ldots, 0, ((-1,-1), \ldots, (-1,-1)))$, a dummy sink node $(n+1)$, and a node $(k_1, \ldots, k_W, grop)$ for each $0 \leq k_w \leq n_w$, $1 \leq w \leq W$ and $grop$ being a general ROP. Node $(k_1, \ldots, k_W, grop)$ represents the first k_w, $1 \leq w \leq W$, aircraft of class w being scheduled and runways being occupied as specified by $grop$.

We say that general ROP $((O_1, w_1), \ldots, (O_R, w_R))$ can be transformed to general ROP $((O'_1, w'_1), \ldots, (O'_R, w'_R))$ with respect to aircraft $a \in A$ if there is exactly one r, $1 \leq r \leq R$, such that

- $(O'_{r'}, w'_{r'}) = (O_{r'}, w_{r'})$ for each r', $r' \neq r$,
- $w'_r = w(a)$,
- if $O_r \geq 0$, then $\max\left\{O_r + s_{w_r,w(a)}, E_a\right\} \leq O'_r \leq L_a$, and
- if $O_r = -1$, then $\max\left\{d_{w(a)}, E_a\right\} \leq O'_r \leq L_a$ where d_w is the duration an aircraft of type w occupies a runway for landing.

We denote $grop \overset{a}{\Rightarrow} grop'$ if general ROP $grop$ can be transformed into general ROP $grop'$ with respect to aircraft a.

We have two types of arcs. First, for two nodes $(k_1, \ldots, k_W, grop)$ and $(k'_1, \ldots, k'_W, grop')$ there exists an arc $((k_1, \ldots, k_W, grop), (k'_1, \ldots, k'_W, grop'))$ if there is exactly one w, $1 \leq w \leq W$, such that

- $k'_{w'} = k_{w'}$ for each w', $w' \neq w$,
- $k'_w = k_w + 1$, and
- $grop \overset{a}{\Rightarrow} grop'$ where a is the k_wth aircraft in A_w.

Arc $((k_1, \ldots, k_W, grop), (k'_1, \ldots, k'_W, grop'))$ has a weight equaling the cost of landing the k_wth aircraft in A_w at the time indicated by $grop'$. Note that there is exactly one r, $1 \leq r \leq R$, such that $(O_r, w_r) \neq (O'_r, w'_r)$. Then O'_r is the landing time of aircraft a.

Second, there is an arc $((k_1, \ldots, k_W, grop), (n+1))$ for each node $(k_1, \ldots, k_W, grop)$ where $k_w = n_w$ for each w, $1 \leq w \leq W$. Weights of these arcs are zero.

Clearly, a path from $(0, \ldots, 0, ((-1,-1), \ldots, (-1,-1)))$ to $(n+1)$ represents a feasible schedule and the shortest path outlines the minimum cost schedule. The constructed graph is acyclic due to the definition of transformability of one general ROP into an other.

We have $O\left(n^W W^R K^R n^{R(W^2+1)}\right) = O(W^R K^R n^{RW^2+W+R})$ nodes. Each node is connected to no more than $O(WR|\mathcal{T}'|) = O\left(WRKn^{W^2+1}\right)$ other nodes. Therefore, we can find the optimal schedule in $O\left(RW^{R+1}K^{R+1}n^{(R+1)W^2+W+R+1}\right)$ time. □

Finally, we will show how to generalize the algorithm as described above to the case where separation times must be considered not only between consecutive aircraft but also between each pair of aircraft landing on the same runway.

Suppose we have a partial schedule, that is, we have scheduled a subset of aircraft in ELW order for each class and in increasing landing times on each runway just as in the algorithm proposed above. It is crucial to observe that among all aircraft of type w landing on runway r only the last one is relevant to consider separation times for an aircraft which is supposed to land next on r. Thus, we can maintain the landing time $C_{w,r}$ of the last aircraft of type w landing on runway r for each $1 \le w \le W$ and $1 \le r \le R$. Having $C_{w,r}$ we can easily compute the earliest possible landing time for the next aircraft which is the only difference of the generalized algorithm to the one proposed above. Clearly, $C_{w,r} \in \mathcal{T}'$ and it is easy to see that the number of nodes to be considered is still polynomial if W and R are fixed.

4 Enhancing an IP model using optimality properties

We provide polynomial time algorithms for the ALP in Section 3. However, due to the large state space we expect the proposed approaches to require excessive computation times. Standard integer programming (IP) solvers (most likely) cannot guarantee polynomial run times for the problems. Still, due to efficient implementation they may provide optimal solutions within reasonable time in the average case. Therefore, in this section we exemplarily incorporate one of the developed optimality properties in an IP model to speed up the solution procedure.

We introduce the traditional binary programming formulation for the ALP and extend it using the ELW property according to the landing time window order within one class of Lemma 1. Based on Beasley et al. (2000) the traditional ALP is modeled as follows. Based on the breakpoint $k^*_{w(a)}$ with zero cost for aircraft a, let $h_{w(a),k}$ be the additional increase of the objective function after breakpoint $k \ge k^*_{w(a)}$ and let $g_{w(a),k}$ be the respective increase before breakpoint $k \le k^*_{w(a)}$. We use the following sets of decision variables:

Continuous decision variables:

C_a assigned landing time for aircraft $a \in A$

$\alpha_{a,k}$ amount of time that the deviation from the target time is smaller than breakpoint $b_k^{w(a)}$; $a \in A, k \le k^*_{w(a)}$. That is $\alpha_{a,k} = \max\{0, T_a - C_a + b_k^{w(a)}\}$.

$\beta_{a,k}$ amount of time that the deviation from the target time is larger than breakpoint $b_k^{w(a)}$; $a \in A, k \ge k^*_{w(a)}$. That is $\beta_{a,k} = \max\{0, C_a - T_a - b_k^{w(a)}\}$.

Binary decision variables:

$$\delta_{aa'} = \begin{cases} 1 & \text{if aircraft } a \text{ lands before aircraft } a' \\ 0 & \text{otherwise} \end{cases} \quad \text{for } a \ne a' \in A$$

$$z_{aa'} = \begin{cases} 1 & \text{if aircraft } a \text{ and } a' \text{land on the same runway} \\ 0 & \text{otherwise} \end{cases} \quad \text{for } a \ne a' \in A$$

$$y_{ar} = \begin{cases} 1 & \text{if aircraft } a \text{ lands on runway } r \\ 0 & \text{otherwise} \end{cases} \quad \text{for } a \in A \text{ and } r \in \{1, \ldots, R\}$$

Beasley et al. (2000) classify three disjoint sets of pairs of flights $(a, a') \in A \times A$ based on the overlap of the respective time windows:

- U_1 is the set of pairs (a, a') where a may land before a' and the other way round.

$$U_1 = \{(a, a') \in A \times A | a \ne a' \text{ and } L_a \ge E_{a'} \text{ and } L_{a'} \ge E_a\} \tag{5}$$

- U_2 is the set of pairs (a, a') where a must land before a' and $L_a + S_{aa'} > E_{a'}$, that is, satisfied separation requirement is not implied by landing time windows.

$$U_2 = \{(a, a') \in A \times A | a \ne a' \text{ and } L_a + S_{aa'} > E_{a'} \text{ and } L_a < E_{a'}\} \tag{6}$$

- U_3 is the set of pairs (a, a') where a must land before a' and $L_a + S_{aa'} \le E_{a'}$, that is, satisfied separation requirement is implied by landing time windows.

$$U_3 = \{(a, a') \in A \times A | a \ne a' \text{ and } L_a + S_{aa'} \le E_{a'}\} \tag{7}$$

We have $\delta_{aa'} = 1$ and $\delta_{a'a} = 0$ for each pair $(a, a') \in U_2 \cup U_3$, respectively. A separation requirement does not need to be stated explicitly for a pair $(a, a') \in U_3$. Using these sets the traditional IP with a piecewise linear objective function can be stated as follows:

$$\text{minimize } F = \sum_{a \in A} \left(\sum_{k=1}^{k^*_{w(a)}} g_{w(a),k}\alpha_{a,k} + \sum_{k=k^*_{w(a)}}^{K} h_{w(a),k}\beta_{a,k} \right) \tag{8}$$

subject to the constraints

$$
\begin{aligned}
& E_a \le C_a \le L_a && a \in A; && (9)\\
& \delta_{aa'} + \delta_{a'a} = 1 && (a, a') \in U_1; && (10)\\
& C_a + S_{aa'} z_{aa'} \le C_{a'} + M\delta_{a'a} && (a, a') \in U_1; && (11)\\
& C_a + S_{aa'} z_{aa'} \le C_{a'} && (a, a') \in U_2; && (12)\\
& \alpha_{a,k} \ge T_a - C_a + b_k^{w(a)} && a \in A, k \le k^*_{w(a)}; && (13)\\
& \beta_{a,k} \ge C_a - T_a - b_k^{w(a)} && a \in A, k \ge k^*_{w(a)}; && (14)\\
& \sum_{r=1}^{R} y_{ar} = 1 && a \in A; && (15)\\
& z_{aa'} \ge y_{ar} + y_{a'r} - 1 && a \ne a' \in A; r \in \{1, \ldots, R\}; && (16)
\end{aligned}
$$

The objective function (8) sums up penalty costs for landing before or after the target landing time. Constraint (9) ensures that each aircraft lands within its time window. Either plane a lands before plane a' or vice versa, which is given by constraints (10) for pairs $(a, a') \in U_1$ without a decision about the landing order. Employing a sufficiently large number M the separation requirement for complete separation is modeled with conditions (11) and (12). These equations implement the sequence of completion times. If aircraft a and a' land on different runways, i.e., $z_{aa'} = z_{a'a} = 0$, no separation is required due to the assumption of independent operating runways. Equations (13) and (14) determine the amount of time by which $T_a - C_a$ exceeds and undershoots, respectively, the kth break point. The fact that the objective function contains both α_{ak} and β_{ak} with nonnegative weights g_{ak} and h_{ak} ensures that $\sum_k \alpha_{ak} \cdot \sum_k \beta_{ak} = 0$. For a given solution of the ALP there are $R!$ permutations of runways with identical landing times and hence an identical objective value. To reduce this symmetry in the solution space for the multi-runway case we added a constraint that the first aircraft has to land on the first runway, i.e., $y_{11} = 1$ holds.

Lemma 1 enables us to modify U_1 and U_2. Let $\overline{U_1}$, $\overline{U_2}$, and $\overline{U_3}$ be the respective sets using the inner-class sequence according to the ELW rule of Lemma 1. This eliminates all pairs of aircraft of identical classes from U_1, i.e,

$$\overline{U_1} = \{(a, a') \in U_1 | w(a) \neq w(a')\} \tag{17}$$

For pairs in $U_1 \setminus \overline{U_1}$ the landing order can be considered given. Still, the separation requirements are not fulfilled inherently. Thus, we must add corresponding tuples to U_2, that is, we set

$$\overline{U_2} = U_2 \cup \{(a, a') \in U_1 | a < a' \text{ and } w(a) = w(a')\} \tag{18}$$

The set of pairs where satisfied separation requirements are implied by landing time windows does not change, i.e., $\overline{U_3} = U_3$ holds.

To demonstrate the influence of the ELW rule, we numerically compare formulation (8)–(16) to the one using $\overline{U_1}$, $\overline{U_2}$, and $\overline{U_3}$ instead of U_1, U_2, and U_3. We use two sets of instances published in the OR-Library of Beasley (Beasley et al. 2000) and in Bianco et al. (1999). The triangle inequality (1) holds for separation requirements of any three classes. Therefore, the respective solution spaces for complete separation and successive separation are identical. The aircraft landing problem is modeled in GAMS 22.6 using CPLEX 11.01 on a 1.8 GHz Dual Core machine with 2 GB of RAM.

The instances from the OR-Library of Beasley assume bounded landing time windows $E_a \leq T_a \leq L_a$ but do not report aircraft classes. Based on the given separation requirement for each pair of aircraft, linear costs for delay, and linear costs for earliness, we are able to derive classes for the first seven instances. This results in problem instances with up to 44 aircraft and 2 or 4 classes of aircraft. All these instances could be solved for $|R| \leq 4$ runways within 30 s with both formulations, while the class-dependent formulations have shorter CPU time in almost all cases.

The second set of instances of Bianco et al. (1999) provides realistic cases of the ALP with 30 and 44 aircraft, respectively. It is a special case of ALP where $E_a = T_a$ and $L_a = \infty$ hold. The instances have two and four aircraft classes, respectively. The first three columns of Table 2 give the number of aircraft, the number of considered classes, and the number of runways. The objective value for three different objective functions is given in column 4. The upper part of Table 2 outlines runtimes for the problem where overall delay is to be minimized, as considered in Bianco et al. (1999). The second and the third part show runtimes for the problem variants with a piecewise linear objective function with three break points at 5, 25, and 50 min (second part) and 5, 15, and 30 min (third part), respectively. The basic delay costs are 1 unit per second during the first 5 min. The additional increase of the delay costs are assumed to be $b_{a1} = 1$, $b_{a2} = 3$, and $b_{a3} = 5$, respectively. The aircraft landing problem is modeled in GAMS 22.6. The last columns in Table 2 show the CPU times. The number in brackets shows the time the first optimal solution was found. The set definition according to the ELW rule enables the solver to proof optimality in all instances. The traditional formulation run out of memory (oom) without providing an optimal solution in three cases with a single runway. In these cases the objective value F_{best} of the best feasible solution found is presented in brackets. In the other cases where the traditional formulation could prove optimality the ELW-based formulation is at least as fast as the traditional formulation. Especially for single runway cases the computation times of the formulation with aircraft classes outperform the traditional one.

5 Conclusions and outlook

In this article, we develop algorithms for the ALP. We provide three algorithms concerning special cases of the problem and a fourth algorithm for the general setting. For ALP with a single runway the algorithm is based on blocks of aircraft which land immediately after each other. The algorithms for multiple runways are based on a state representation via runway occupancy profiles. In the general setting, we consider multiple runways, bounded landing time windows, and a piecewise linear cost function with multiple breakpoints for each class of aircraft. Regarding computational complexity of ALP, we can summarize our results as follows. ALP is polynomially solvable if

- the number of runways is fixed,
- the number of aircraft classes is fixed, and
- the number of break points of each class' cost function is polynomial in input size.

Table 2 Results for the examples of Bianco et al. (1999)

$\|A\|$	$\|W\|$	$\|R\|$	F_{opt}	ELW formulation (s)	Traditional formulation (s)
30	4	1	3,721	11 (2)	15,980 (7)
		2	219	3 (3)	3 (3)
44	2	1	20,391	24,423 (8,514)	oom after 11,254 ($F_{best} = 21,151$)
		2	660	196 (28)	604 (29)
30	4	1	4,034	7 (2)	1,539 (226)
		2	219	4 (4)	4 (4)
44	2	1	31,037	20,180 (15,276)	oom after 4,193 ($F_{best} = 34,481$)
		2	660	93 (21)	1,294 (24)
30	4	1	4,034	5 (2)	415 (34)
		2	219	4 (4)	4 (4)
44	2	1	32,864	20,924 (17,878)	oom after 4,808 ($F_{best} = 40,476$)
		2	660	98 (21)	1,363 (27)

This result considerably extends previously existing results. While this result follows from the algorithm designed for the most general setting of ALP the other algorithms employ optimality properties for more special cases to enhance their performance.

While the algorithms cannot be expected to perform sufficiently fast for real world applications we, nevertheless, obtain valuable insights into the problem's structure. These insights may be used to enhance standard solution techniques to solve ALP approximately. We exemplarily show the benefit from incorporating such insights into a common IP model for ALP by conducting a computational study comparing the standard model and our enhanced version. For the analyzed instances, the formulation based on the ELW rule outperforms the traditional one. The presented DP framework could also serve as starting point for efficient (heuristic) solution methods for future research.

Further research could also be directed to add more constraints related to airport-specific air traffic control restrictions. In particular, limiting the number of overtaking manoeuvers as considered in Balakrishnan and Chandran (2010) can be incorporated in our approaches easily since in each state of DP methods, we have the number of aircraft of each class already scheduled and, therefore, can limit the set of feasible states to those where the aircraft currently scheduled takes a feasible position.

It is obvious that the optimality properties, such as ELW rule, block structure, and fixed aircraft, developed in the article at hand give rise to many additional model variants. Investigating a larger variety of IP models in terms of run time behavior of standard solvers seems to be a promising field for further research. Also, as mentioned in Sect. 2.2, there are applications of our problem in other fields such as vehicle routing. Therefore, a second perspective for future research is to generalize our problem focusing on requirements arising in other applications and determining the resulting problem's computational complexity.

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